



Estimation of static formation temperatures in geothermal wells by using an artificial neural network approach

A. Bassam^a, E. Santoyo^{a,*}, J. Andaverde^b, J.A. Hernández^b, O.M. Espinoza-Ojeda^a

^a Centro de Investigación en Energía, Universidad Nacional Autónoma de México, Privada Xochicalco s/n, Centro, 62580 Temixco, Morelos, México

^b Centro de Investigación en Ingeniería y Ciencias Aplicadas, Universidad Autónoma del Estado de Morelos, Av. Universidad 1001, Col. Chamilpa, 62209 Cuernavaca, Morelos, México

ARTICLE INFO

Article history:

Received 13 August 2009

Received in revised form

6 January 2010

Accepted 7 January 2010

Keywords:

Horner method

Geothermal energy

Levenberg–Marquardt algorithm

Borehole drilling

Bottom-hole temperature

Artificial intelligence

Shut-in time

ABSTRACT

An artificial neural network (ANN) approach was used to develop a new predictive model for the calculation of static formation temperature (SFT) in geothermal wells. A three-layer ANN architecture was successfully trained using a geothermal borehole database, which contains “statistically normalised” SFT estimates. These estimates were inferred from seven analytical methods commonly used in geothermal industry. Bottom-hole temperature (BHT) measurements and shut-in times were used as main input variables for the ANN training. Transient temperature gradients were used as secondary variables. The Levenberg–Marquardt (LM) learning algorithm, the hyperbolic tangent sigmoid transfer function and the linear transfer function were used for the ANN optimisation. The best training data set was obtained with an ANN architecture composed by five neurons in the hidden layer, which made possible to predict the SFT with a satisfactory efficiency ($R^2 > 0.95$). A suitable accuracy of the ANN model was achieved with a percentage error less than $\pm 5\%$. The SFTs predicted by the ANN model were statistically analyzed and compared with “true” SFTs measured in synthetic experiments and actual BHT logs collected in geothermal boreholes during long shut-in times. These data sets were processed both to validate the new ANN model and to avoid bias. The SFT estimates inferred from the ANN validation process were in good agreement ($R^2 > 0.95$) with the “true” SFT data reported for synthetic and field experiments. The results suggest that the new ANN model could be used as a practical tool for the reliable prediction of SFT in geothermal wells using BHT and shut-in time as input data only.

© 2010 Elsevier Ltd. All rights reserved.

1. Introduction

The exploitation of geothermal resources for producing electricity requires the drilling of deep boreholes in the most suitable thermal regions of geothermal fields (Saito et al., 1998; Davis and Michaelides, 2009). The borehole drilling is a complex process in which a constant thermal anomaly (added to a circulating drilling mud) affects the original rock-formation temperature surrounding the borehole (Fomin et al., 2003, 2005). Once the borehole drilling is completed, the resulting thermal recovery is evaluated by analyzing build-up bottom-hole temperature (BHT) and shut-in time measurements (Santoyo et al., 2000). BHT data are measured at different shut-in times during the borehole drilling operations (Espinoza-Paredes and Espinoza-Martinez, 2009). BHTs are usually costly due to the use of sophisticated logging equipment, but mostly, because the borehole drilling progress needs to be stopped (Wisian et al., 1998; Fomin et al., 2005).

The determination of the static formation temperatures (SFTs) from BHT measurements constitutes a crucial task for the evaluation of geothermal systems (Espinoza et al., 2001; Kutasov and Eppelbaum, 2003). The estimation of SFT from BHT data at early drilling times offers the chance to know the virgin formation temperature several months before it can be measured with accuracy. This approach is valuable in terms of planning, exploration, developing, and exploitation programmes. The knowledge of the SFT is required for the: (1) determination of geothermal gradients for exploration mapping; (2) determination of heat flows of promising geothermal zones; (3) interpretation of borehole logging; (4) optimal design of borehole drilling and completion programs; (5) location of permeable or lost circulation zones; (6) evaluation of *in-situ* thermophysical formation properties; and (7) for the estimation of the heat reserves in geothermal systems.

The estimation of SFT in geothermal boreholes is usually performed by using simplified analytical methods based on complex heat transfer models concentrated at the bottom-hole conditions, where BHTs are actually measured. The BHT measurements tend to reflect the thermal anomalies caused by the drilling mud circulation to the rock-formation.

* Corresponding author.

E-mail address: esg@cie.unam.mx (E. Santoyo).

The most common analytical methods include: (i) the radial source with a conductive heat flow or Brennand method (BM: Brennand, 1984); (ii) the cylindrical heat source with a conductive-convective heat flow method (CSM) proposed by Hasan and Kabir (1994); (iii) the line-source or the well-known Horner-plot method (HM: Dowdle and Cobb, 1975); (iv) the generalized Horner or the Kutasov–Eppelbaum method (KEM: Kutasov and Eppelbaum, 2005); (v) the cylindrical source with a conductive heat flow or Leblanc method (LM: Leblanc et al., 1981); (vi) the cylindrical source with a conductive heat flow or Manetti method (MM: Manetti, 1973); (vii) the spherical and radial heat flow method (SRM) proposed by Ascencio et al. (1994); among others.

Most of these analytical methods require at least three or more BHT measurements carried out at the same borehole depth but at different shut-in times. The SFT is determined through the mathematical solution of each analytical method, using BHT and shut-in time measurements as input data, and the linear or nonlinear regressions between BHT data and the time functions of each method (Verma et al., 2006a, b).

Although considerable progress has been achieved in this area over recent years, large discrepancies among the SFT results provided by different methods have been reported (Andaverde et al., 2005). Therefore, the development of new reliable methods to estimate SFT is still a challenging area. The geothermal industry is interested in estimating SFT in systems where heat convection dominates. Heat convection is attributed to the presence of permeable zones in reservoirs and drilling mud losses into the formation. The determination of SFT under such conditions requires the use of complex analytical methods or wellbore simulators. These tools need thermophysical and transport properties of drilling, cementing, and formation materials, which are rarely available in the literature. In addition, the accurate knowledge of the mud circulation time, which is generally unknown or difficult to determine under drilling conditions, is also needed by most analytical methods (Hermanrud et al., 1990).

Considering this complex scenario, the geothermal industry requires new practical tools that use the commonly available data (i.e., BHT and shut-in time) to estimate the SFT with an acceptable accuracy. In this regard, a new application of artificial neural networks (ANN) is here proposed for developing a new reliable method for the calculation of SFTs in geothermal wells.

ANN have been suggested as powerful computational tools in many sciences for modelling and solving complex real-world problems, and mainly for forecasting applications (Zhang et al., 1998). The use of ANN is increasingly in Earth Sciences (e.g., Goutorbe et al., 2006; Hsieh et al., 2009; Leite and Filho, 2009a, b; Morton, 2009). In recent years, the application of ANN to geothermal and petroleum engineering problems has also been the subject of study (e.g., Farshad et al., 2000; Bayram, 2001; Can, 2002; Yilmaz et al., 2002; Spichak and Goidina, 2005; Spichak, 2006; Díaz-González et al., 2008; Serpen et al., 2009).

In this work, a three-layer ANN approach was successfully trained for obtaining a new predictive model to estimate SFT in geothermal wells. The new ANN model requires BHT and shut-in time measurements (as main input data), normalised SFT estimates (inferred from seven analytical methods), and transient temperature gradients.

2. Methodology

The ANN model was trained with a geothermal borehole database containing SFT estimates (statistically normalised). The geothermal database also contains build-up thermal recovery logs recorded during drilling operations carried out in some geothermal boreholes of the world. Build-up thermal recovery logs

consist of BHT and shut-in time measurements, and transient temperature gradients, together with SFT estimates inferred from the regression analysis of curvilinear relationships between BHT and shut-in time data using simplified analytical methods. SFT estimates were calculated with seven analytical methods commonly used in geothermal industry.

The normalisation of SFT estimates was carried out using statistical tests for discordant outlier detection and rejection (Verma, 2005). Mean SFT estimates (normalised) and standard deviations were computed using an improved outlier detection/rejection statistical method (Verma et al., 2009). This statistical approach coupled to the ANN enables the resulting errors to be effectively detected and rejected before selecting reliable SFT estimates. The new predictive ANN model was evaluated by using four BHT data sets (two synthetic and two actual borehole data sets) measured in experimental and long logging works where the SFT or the true formation temperature (TFT) was already known. These validation tests were considered as an accuracy evaluation of the ANN model.

2.1. Geothermal borehole databases for learning and validation tests

A geothermal borehole database containing BHT and shut-in time (Δt) measurements logged from drilling operations of 10 boreholes was created (Table 1). These thermal recovery data sets were collected from the reports of boreholes drilled in various world geothermal fields: (1) Los Humeros, Mexico [MXCO1, number of BHT measurements (m_{BHT})=6; shut-in times (Δt) up to 36 h after cessation of drilling mud circulation (Andaverde et al., 2005)]; (2) Los Humeros, Mexico [MXCO2, m_{BHT} =7; $\Delta t \leq 42$ h (Andaverde et al., 2005)]; (3) Oklahoma, USA [SGIL, m_{BHT} =13; $\Delta t \leq 12$ h (Schoepel and Gilarranz, 1966)]; (4) Mississippi wellbore, USA [USAM, m_{BHT} =14; $\Delta t \leq 200$ h (Kutasov, 1999)]; (5) Leyte, Philippines [PHIL, m_{BHT} =14; $\Delta t \leq 15.58$ h (Andaverde et al., 2005)]; (6) Kyushu, Japan [JAPN, m_{BHT} =10; $\Delta t \leq 72.5$ h (Hyodo and Takasugi, 1995)]; (7) Chipilapa, El Salvador [CH-A1, m_{BHT} =5; $\Delta t \leq 95.27$ h (González-Partida et al., 1997)]; (8) Chipilapa, El Salvador [CH-A2, m_{BHT} =6; $\Delta t \leq 190.5$ h (Iglesias et al., 1995)]; (9) Larderello, Italy [ITAL, m_{BHT} =40; $\Delta t \leq 27$ h (Da-Xin, 1986)]; and (10) Norton Sound field, Alaska [COST, m_{BHT} =6; $\Delta t \leq 75.5$ h (Cao et al., 1988a)].

Four thermal recovery data sets were additionally compiled for the validation or accuracy tests:

(i) two synthetic data sets collected from experimental works reported in the literature (SHBE, m_{BHT} =8: Shen and Beck, 1986; and CLAH, m_{BHT} =15: Cao et al., 1988b); and (ii) two data sets logged in some geothermal boreholes which were collected during long logging works (CH-A4 borehole, m_{BHT} =5: Iglesias et al., 1995; and KJ-21 borehole, m_{BHT} =6: Steingrímsson and Gudmundsson, 2006). These data sets were considered for the evaluation of accuracy because the TFTs were already known (SHBE=80.0 °C; CLAH=120.0 °C; CH-A4=169 °C; and KJ-21=240 °C, respectively).

2.2. Initial estimation of SFTs using analytical methods

Seven analytical methods based on complex heat transfer models were used for the initial calculation of SFTs. The seven analytical methods were: (i) the Horner-plot method, HM; (ii) the Brennand method, BM; (iii) the Kutasov–Eppelbaum method, KEM; (iv) the Leblanc method (LM); (v) the Manetti method (MM); (vi) the conductive spherical-radial method (SRM); and (vii) the conductive-convective cylindrical source method (CSM). The simplified solutions of these analytical methods are summarised in Table 2. SFTs were estimated using conventional

Table 1

Thermal recovery data sets used for ANN training and validation.

Data set	Borehole data for training the new ANN model						Synthetic data for the validation					
	<i>t</i> (h)	BHT (°C)	Data set	<i>t</i> (h)	BHT (°C)	Data set	<i>t</i> (h)	BHT (°C)	Data set	<i>t</i> (h)	BHT (°C)	
MXCO1 <i>t_c</i> =2.5 h <i>m</i> =6	6	186.3	PHIL <i>t_c</i> =15 h <i>m</i> =14	2.58	93	ITAL <i>t_c</i> =5 h <i>m</i> =40	5.7	84.5	SHBE <i>t_c</i> =5 h <i>m</i> =8	2.5	56.6	
	12	206.9		3.58	88		6.0	93.1		5	61.3	
	18	219.0		4.58	99		6.5	95.0		7.5	64.3	
	24	231.9		5.58	108		7.0	96.2		10	66.6	
	30	239.8		6.58	112		7.5	97.5		15	69.6	
	36	247.7		7.58	117		8.0	98.4		20	71.7	
MXCO2 <i>t_c</i> =2.5 h <i>m</i> =7				8.58	120		8.5	99.3		30	74.1	
	6	178.6		9.58	126		9.0	100.3		40	75.5	
	12	198.9		10.58	133		9.5	101.1				
	18	211.4		11.58	133		10.0	102.0				
	24	225.8		12.58	134		10.5	102.7				
	30	235.1		13.58	137		11.0	103.4	CLAH <i>t_c</i> =5 h <i>m</i> =15	2	91.7	
	36	240.4		14.58	141		11.5	104.2		4	98.1	
	42	247.1		15.58	146		12.0	104.7		6	102.4	
						12.5	105.4	8		105.5		
						13.0	106.1	10		107.9		
						13.5	106.7	12		109.8		
SGIL <i>t_c</i> =3 h <i>m</i> =13	2	86.56	JAPN <i>t_c</i> =5 h <i>m</i> =10	7.5	103.0		14.0	107.6		14	111.3	
	3	88.84		8.5	107.4		14.5	108.3		16	112.5	
	4	90.76		9.5	110.4		15.0	108.9		18	113.6	
	5	92.29		12.5	119.6		15.5	109.6		20	114.4	
	6	93.44		15.5	126.6		16.0	110.2		22	115.2	
	7	94.21		18.5	132.8		16.5	110.8		24	115.8	
	8	94.59	24.5	142.4		17.0	111.3		30	117.1		
	9	94.98	72.5	170.9		17.5	111.9		40	118.4		
	10	95.36				18.0	112.4		50	119.1		
	11	95.74	CH-A1 <i>t_c</i> =5 h <i>m</i> = 5	6	90		18.5	112.9				
	12	96.13		12	102		19.0	113.3	CH-A4 <i>t_c</i> =5 h <i>m</i> =5	6	85	
				26.57	116		19.5	113.8		12	94	
				47.1	130		20.0	114.2		26.57	122	
USAM <i>t_c</i> =5 h <i>m</i> =14				95.27	145		20.5	114.7			47.1	139
	1	138.75					21.0	115.1		95.27	152	
	2	139.55					21.5	115.4				
	5	140.89	CH-A2 <i>t_c</i> =5 h <i>m</i> =6	6.0	231		22.0	115.7	KJ-21 <i>t_c</i> =2.5 h <i>m</i> =6	2	50	
	10	142.03		12.0	231		22.5	116.1		30	125	
	15	142.72		26.6	235		23.0	116.5		52	158	
	20	143.21		47.1	237		24.0	117.1		96	184	
	25	143.60		95.3	239		25.0	117.7		242	218	
	30	143.92		190.5	240		26.0	117.9		556	225	
	40	144.42					27.0	118.7				
	50	144.81	COST <i>t_c</i> =4 h <i>m</i> =6	8.5	45.00							
	70	145.40		10.0	49.44							
	100	146.04		16.5	55.56							
	150	146.76		34.5	56.11							
200	147.27	38.0		57.22								
		75.5		56.11								

t_c—drilling mud circulation time; additional data reported for each data set: SGIL ($R=0.10028$, $k=2.25419$ W/m °C, $c=878.64$ J/kg °C, and $\rho=2306.6$ kg/m³); COST ($k=4.97$ W/m °C); SHBE ($r=0.108$ m, TFT=80 °C); CLAH ($r=0.108$ m, TFT=120 °C); CH-A4 (TFT=169 °C); KJ-21 (TFT=240 °C). *m*—number of data.

Table 2

Listing of major numerical solutions of analytical methods (HM, BM, KEM, LM, MM, SRM, and CSM) used for determining SFT in geothermal boreholes.

Analytical method	Numerical solution
BM	$BHT(t) = T_{BM} - b_{BM} \left(\frac{1}{\Delta t + p_{LC}} \right)$
CSM	$BHT = T_{CSM} + (b_{CSM})[F(t_{CD} + \Delta t_D) - F(t_D)]$
HM	$BHT(t) = T_{HM} + (b_{HM}) \ln((t_c + \Delta t) / \Delta t)$
KEM	$BHT(t) = T_{KEM} + b_{KEM} \ln(X)$
LM	$BHT(t) = T_{LM} - b_{LM} \left[1 - \exp \left(- \frac{r_{LC}^2}{4\alpha \Delta t} \right) \right]$
MM	$BHT(t) \approx T_{MM} + b_{MM} \ln \left(\frac{\Delta t}{\Delta t - t_c} \right)$
SRM	$BHT(t) = T_{SRM} + (b_{SRM}) \left(\frac{1}{\Delta t} \right)$

BHT: Bottom-hole temperature; Δt : shut-in time; SFTs were inferred from the intercept value of each analytical method: T_{BM} , T_{CSM} , T_{HM} , T_{KEM} , T_{LM} , T_{MM} , and T_{SRM} . For more details of primary and secondary variables, the readers are referred to the original sources of the methods (indicated in the introduction section).

linear (ordinary and weighted) and quadratic (QR) regressions between the BHT and the time function data of each method (see Andaverde et al., 2005). Ordinary linear regressions (OLR) neglect the errors in the *x* (time functions) and *y* (BHT) variables for the calculation of the regression parameters: intercept (*a*) and slope (*b*), whereas the weighted linear regressions (WLR) consider the errors of *x* and *y* (*s_x* and *s_y*) for the estimation of the regression parameters (Verma, 2005).

According to the numerical procedure used by each analytical method, SFTs were inferred from the intercept values (*a*) of the OLR, WLR and QR (e.g., Santoyo et al., 2000; Andaverde et al., 2005).

2.3. General scheme for the SFT estimation

A schematic flow diagram showing the training procedure used by the ANN model is shown in Fig. 1. Five fundamental

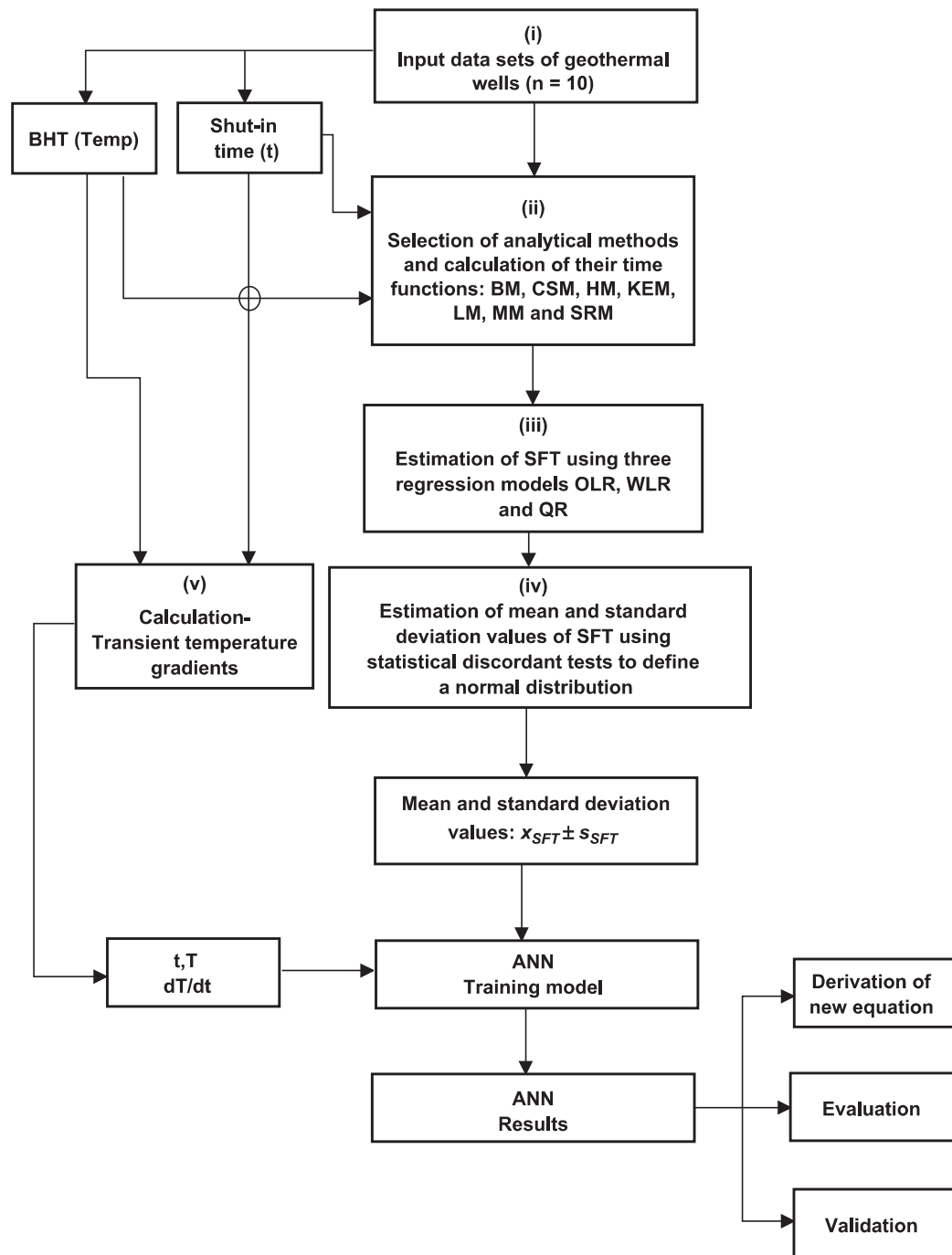


Fig. 1. Schematic flow diagram of ANN model used for determining SFT in geothermal boreholes.

sections were included in the training process: (i) the input data reading ($j=10$, containing 120 pairs of BHT and shut-in time data); (ii) the selection of analytical methods and calculation of their time functions per data set; (iii) the determination of the SFT by seven analytical methods using three different regression models: OLR, WLR, and QR per data set; (iv) the application of statistical discordant tests to calculate the mean values of the SFT estimates and their standard deviations per data set; and (v) the calculation of temperature gradients using the transient thermal recovery data of the boreholes.

SFT estimates were individually computed by using the 10 data sets (MXCO1, MXCO2, SGIL, USAM, PHIL, JAPN, CH-A1, CH-A2, ITAL, and COST), which contain BHT and shut-in time data. The time functions of each analytical method (HM, BM, KEM, LM, MM, SRM,

and CSM) were calculated by means of their corresponding equations (Table 2). Three SFT estimates were individually inferred from the regression analysis (OLR, WLR, and QR) using each analytical method and referred to each data set. A total number of 21 SFT estimates per data set were finally computed.

For training and correlating the borehole thermal recovery data sets with reliable SFT values, mean values of the 21 SFT estimates were computed per data set through a statistical normalisation using the software DODESYS (Verma et al., 2009). Nine discordant tests (N1, N2, N4, N7, N8, N9, N10, N14, and N15 proposed by several authors: see Verma, 2005) were applied to the 21 SFT estimates of each data set; firstly, to identify and to reject discordant outliers, and secondly, to calculate the mean SFT ($\bar{x}_{SFT}$) and its associated standard deviation (s_{SFT}).

This method of statistical normalisation enabled the large resulting errors derived from the analytical methods to be efficiently detected and rejected before selecting the best SFT estimates. Table 3 summarises the results obtained after using the statistical normalisation procedure.

The mean SFT estimates and discordant outliers detected after applying the statistical tests are plotted in Fig. 2. Thirty discordant outliers were detected and rejected. These outliers corresponded to the SFT estimates inferred from the SRM which systematically overestimate the SFT (Andaverde et al., 2005). The mean values of the SFT estimates ($\bar{x}_{SFT}$) and their standard deviations (s_{SFT}) per data set were used for the ANN training. Three SFT values were finally employed for transporting the uncertainties (the mean value: $\bar{x}_{SFT}$; the mean value plus the standard deviation: $\bar{x}_{SFT} + s_{SFT}$; and the mean value minus the standard deviation: $\bar{x}_{SFT} - s_{SFT}$) (see Table 3).

Transient temperature gradients were computed using the following equation:

$$\frac{dT}{dt} = \frac{(BHT_{i+1} - BHT_i)}{t_{i+1} - t_i} \quad (1)$$

Table 3

Mean SFT estimates inferred from 21 SFT predictions of seven analytical methods (using three different regression models OLR, WLR, and QR), and after applying a statistical normalisation process.

Data set	SFT (analytical methods) $\bar{x}_{SFT} \pm s_{SFT}$	SFT (predicted—ANN) $\bar{x}_{ANN} \pm s_{ANN}$
MXCO1	263 ± 15	256 ± 20
MXCO2	262 ± 18	253 ± 15
USAM	146 ± 1	139 ± 5
ITAL	132 ± 10	133 ± 8
PHIL	197 ± 28	185 ± 25
JAPN	176 ± 18	180 ± 11
CH-A1	147 ± 5	151 ± 7
CH-A2	241 ± 18	243 ± 15
COST	57 ± 3	56 ± 2
SGIL	100 ± 3	102 ± 9

where BHT denotes the bottom-hole temperatures, t the shut-in times, and the subscripts i provides the transient values of these variables.

The transient temperature gradient (dT/dt) was an important parameter for training the ANN. When (dT/dt) tend to zero at infinite time ($t \rightarrow \infty$), it means that the geothermal formation temperature approaches to thermal equilibrium, and therefore:

$$\lim_{t \rightarrow \infty} \frac{dT}{dt} = 0 \therefore (BHT \approx SFT) \quad (2)$$

The SFT_{ANN} was defined as the unique output parameter of the ANN model.

2.4. ANN architecture

The neurons are grouped into distinct layers and interconnected according to a given computational architecture. The network function is largely determined by the connections between neurons. Each connection among two neurons has a weighting coefficient. The standard network structure for an approximation function is the multiple-layer perception (or the well-known feed forward network). The feed forward network often has one or more hidden layers of sigmoidal neurons followed by an output layer of linear neurons. Multiple-layers of neurons with nonlinear transfer functions allow the network to learn nonlinear and linear relationships between input and output vectors. The linear output layer enables the network to generate values outside the -1 to $+1$ range (e.g., Limin, 1995). For the network, the appropriate notation is used in two-layer networks (e.g. Demuth and Beale, 1998). A simplified sketch of the network structure is schematically represented in Fig. 3. The number of neurons in the input and output layers is, respectively, given by the number of input and output variables of the process under investigation. In this work, the input layer can consist of a set of different variables, In_k (i.e., BHT , shut-in time, and dT/dt), while the output layer can contain the output or target variables of the model, Out_l (i.e., SFT), see Fig. 3.

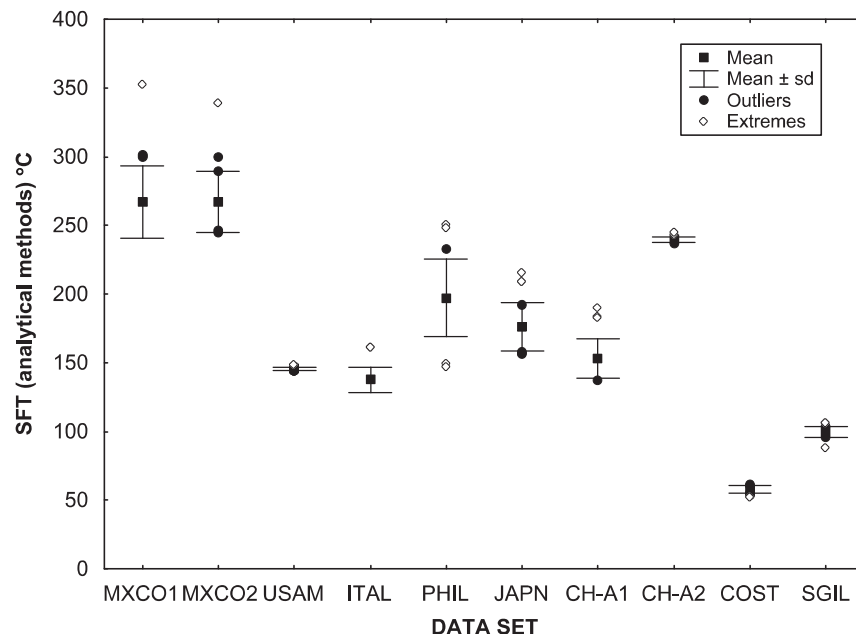


Fig. 2. Mean SFT estimates obtained from normal statistical samples after applying nine statistical discordant tests. Mean SFT estimates per data set were: MXCO1: 263 ± 15 °C; MXCO2: 262 ± 18 °C; USAM: 146 ± 1 °C; ITAL: 132 ± 10 °C; PHIL: 197 ± 28 °C; JAPN: 176 ± 18 °C; CH-A1: 147 ± 5 °C; CH-A2: 241 ± 18 °C; COST: 57 ± 3 °C; and SGIL: 100 ± 3 °C.

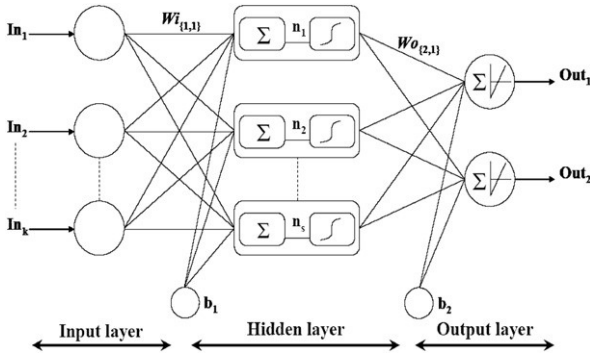


Fig. 3. General ANN architecture (k =input number; In=Input; Out=Output; Continuous lines=weights and bias).

The optimal number of neurons in the hidden layer (S) is difficult to specify, and it generally depends on the problem complexity. This number (S) is iteratively determined. Each neuron in the hidden layer has a bias, $b_1(s)$, which is added to the sum of the product (weights multiplied by its corresponding input variables). In general, the optimal neuron (n_s), is given by the following equation:

$$n_s = W_{i[s,1]}In_1 + W_{i[s,2]}In_2 + \dots + W_{i[s,k]}In_k + b_1(s) \quad (3)$$

The coefficients associated with the hidden layer are grouped into a matrix W_i (given by weights) and the vector $b_1(s)$ of biases. The output layer computes the weighted sum of the signals provided by the hidden layer, and the associated coefficients are grouped into a matrix W_o and the vector $b_2(l)$. Using the matrix notation, the network output can be expressed by the following general equation:

$$Out = g(W_o f(W_i In + b_1) + b_2) \quad (4)$$

Hidden layer neurons may use any differentiable transfer function to generate their output. In this work, a hyperbolic tangent sigmoidal transfer function ($f=tansig$) and a linear transfer function ($g=purelin$) were used. As a result, Eq. (4) can be re-written using these transfer functions to provide the following general equation:

$$Out_{(l)} = \sum_{s=1}^S \left[W_o(l,s) \left(\frac{2}{1 + \exp \left(-2 \left(\sum_{k=1}^K (W_i(s,k)In(k)) + b_1(s) \right) \right)} - 1 \right) \right] + b_{2(l)} \quad (5)$$

where K , S , and l are the number of input variables, the number of neurons in the hidden layer, and the number of the output variables in the ANN architecture, respectively.

3. Neural network learning and testing

A learning (or training) algorithm is defined as a process that consists of adjusting the coefficients (weights and biases) of a network, to minimize an error function (usually a quadratic one) between the network outputs (for a given set of inputs) and the correct outputs (already known). If smooth non-linearities are used, the error function gradient can be computed by the classical back propagation model (Rumelhart et al., 1986).

Previous learning algorithms have used this gradient directly in a steepest descent optimisation, but recent results have shown that second order methods are far more efficient. In this work, the Levenberg–Marquardt (LM) optimisation algorithm (described in the Matlab Neural Network Toolbox) was used (Demuth and Beale, 1998). This algorithm is basically an approximation of

Newton's method (Eq. (6)), which is suggested to approach a second order training speed without having to compute the Hessian matrix (Hagan and Menhaj, 1994)

$$x_{k+1} = x_k - [J^T J + \mu I]^{-1} J^T e \quad (6)$$

where J is the Jacobian matrix that contains first derivatives of the network errors with respect to the weights and biases; e is the vector of network errors; $J^T J$ is an approximation of the Hessian matrix; and $J^T e$ is the gradient. When the scalar μ is zero, Eq. (6) becomes the classical Newton's method using the approximate Hessian matrix. On the other hand, when μ is large, Eq. (6) becomes a more efficient descent gradient with a small step size. The Jacobian matrix can be computed through a standard back-propagation technique (see Hagan and Menhaj, 1994).

Despite the fact that computations involved in iterations were more complex than the steepest descent case, the convergence was faster (typically by a factor of 100). The root mean square error (RMSE) was calculated using the experimental and simulated ANN data (Eq. (7)). This calculation was used as an optimisation criterion for the model adequacy (Fig. 4).

The experimental database (see Table 1) were split into learning (60% of experimental data set; $n=180$) and testing (40% of experimental data set; $n=120$) databases to obtain a statistically representative sample of the problem

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Exp(i) - Sim(i))^2} \quad (7)$$

3.1. ANN training

The new neural network model trained for this work (Fig. 5) finally involved five neurons ($n_s=5$) in the hidden layer with 20 weights (15 elements of a matrix W_i and 5 elements of the matrix W_o), and 6 biases ($b_1=5$ and $b_2=1$) to calculate a SFT value

$$SFT = \sum_{j=1}^S \left[W_o(1,j) \left(\frac{2}{1 + \exp \left(-2 \left(\sum_{k=1}^K (W_i(j,k)In(k)) + b_1(j) \right) \right)} - 1 \right) \right] + b_2 \quad (8)$$

where S and K were already defined and $l=1$ (see Table 4).

During the learning process, the algorithm of the ANN model was evaluated by examining the RMSE value per iteration, together with a different number of neurons in the hidden layer (ranging from one to five neurons). The obtained results (data not shown) proved that the typical learning error decreased when the number of neurons in the hidden layer increased. This observation was evident because of the number of adjusted parameters increased. Nevertheless, one of the problems that could occur during feed-forward neural network training is the well-known “over-fitting” (e.g. Hernández-Pérez et al., 2004).

The comparison of the RMSE calculated for the learning and testing databases was a good criterion to optimize the number of iterations, and to avoid the over-fitting problem. In this neural network, the RMSE showed that for six neurons in the hidden layer, the learning database value was too small with respect to the testing database.

According to RMSE results, the optimal number of neurons in the hidden layer was fixed as five neurons (Fig. 5). The predicted SFT values computed with the ANN model together with their uncertainties are also included in Table 3. A good agreement between the SFT estimates (inferred from analytical methods: x-axis) and the simulated data, SFT_{ANN} (obtained from the ANN model: y-axis) was found (Fig. 6). It can be observed that the

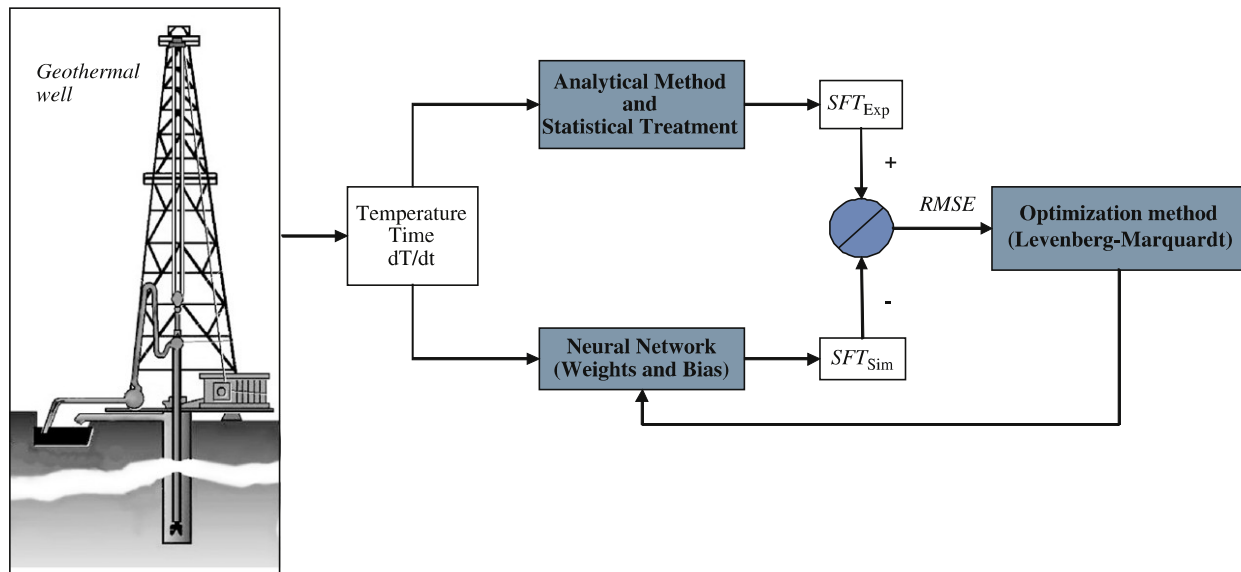


Fig. 4. Numerical algorithm of ANN used for learning and optimizing processes.

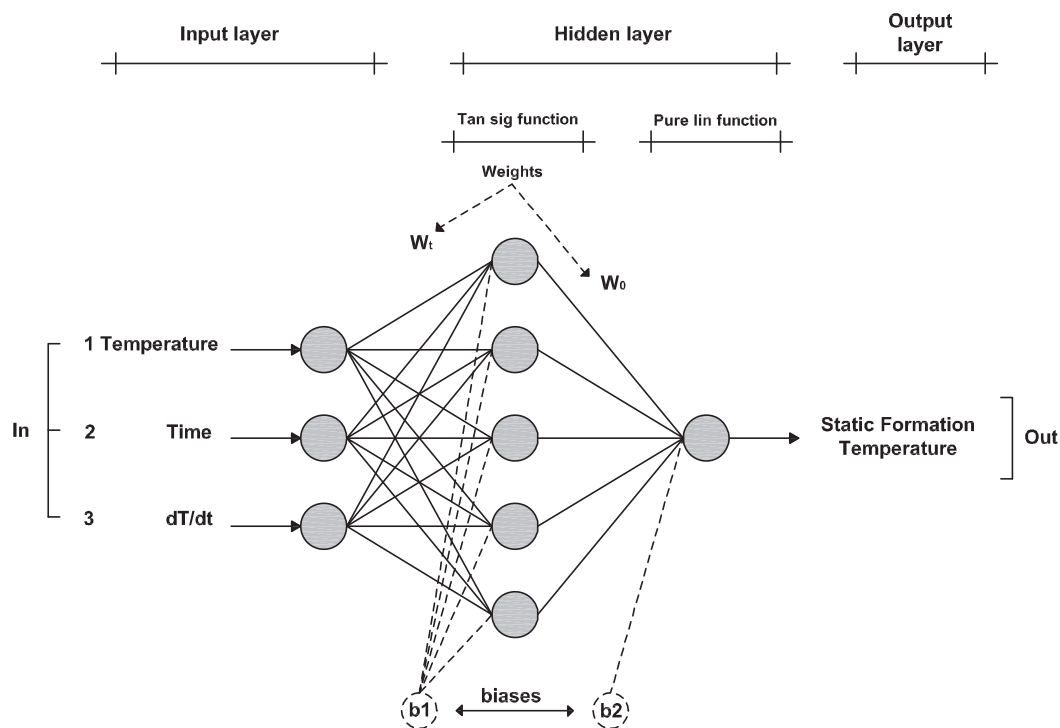


Fig. 5. ANN architecture used for estimating SFT in geothermal boreholes.

simulated SFT_{ANN} data exhibit a good linear relationship with respect to those estimates inferred from analytical methods ($y=0.94376x+8.5631$; $R^2=0.95$). This linear relationship was also confirmed by analyzing the small residuals computed between the measured and predicted SFT values.

The optimum weighting coefficients (W_i , W_o , b_1 , and b_2) obtained for the ANN model using five neurons in the hidden layer are reported in Table 4. These coefficients were used to validate the ANN model with testing data.

3.2. ANN testing

The new ANN model (Eq. (8)) was validated by using the testing data reported for synthetic experiments (SHBE: TFT=80 °C; and CLAH: true TFT=120 °C); and for actual geothermal borehole measurements (CH-A4: TFT=169 °C; and KJ-21: true TFT=240 °C) (see Table 1). The results from these four accuracy tests showed a good agreement (less than a $\pm 5\%$) between the measured SFT (also called TFT) and predicted SFT data (Fig. 7).

Table 4
Optimised weighting and bias parameters of new ANN model.

ANN parameters	With $l=1$, $S=5$ and $K=3$				
W_i	3.9757	1.4220	−2.9996		
	5.0716	13.9674	−2.3583		
	−6.9071	−2.7251	−34.0318		
	−6.3881	2.3519	−6.0095		
	−5.2657	3.6209	−4.5029		
W_o	−1.4449	0.1627	−0.3257	0.4320	−0.5981
b_1			−5.6937		
			−10.0967		
			1.6998		
			−0.5554		
			−2.6091		
b_2			−1.4001		

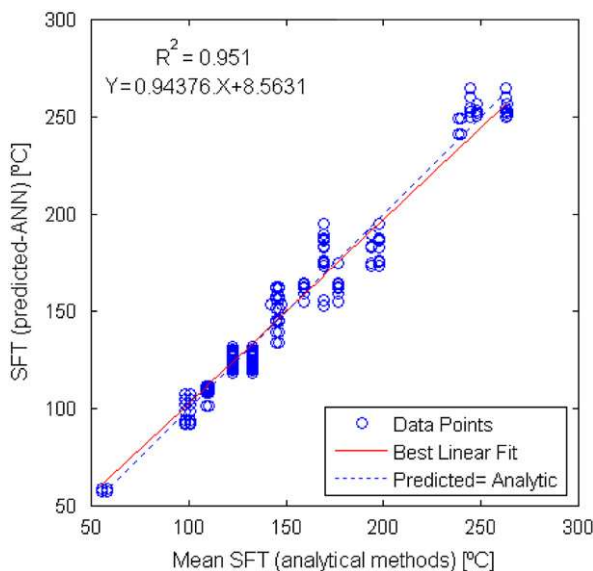


Fig. 6. Statistical comparison between SFT estimates (inferred from analytical methods) and ANN simulated SFT values.

For the synthetic data sets, the ANN model predicted SFT values ranging from 74.5 to 83.4 °C, with a mean SFT_{ANN} value of 79.1 °C for the SHBE data set; and SFT values in the interval 117.8–125.6 °C, with a mean SFT_{ANN} value of 122.1 °C for the CLAH data set. Whereas for the borehole data sets: CH-A4 and KJ-21, the ANN model predicted SFT values in the interval 167.6–173.5 °C, with a mean SFT_{ANN} value of 171.2 °C; and SFT values ranging from 239.1 to 247.1 °C, with a mean SFT_{ANN} value of 243.3 °C, respectively.

From these results, it can be reliably demonstrated that the new ANN model (given by the Eq. (8)) provides acceptable estimates of the SFT with typical errors commonly reported in artificial intelligence engineering applications (e.g. Leite and Filho, 2009a).

The small residuals calculated with the testing database, as well as the good regression coefficient obtained ($R^2=0.95$), enabled to confirm the satisfactory agreement between the measured and predicted SFT data (see Fig. 7). The ANN model developed was therefore able to predict the SFT throughout the experimental data domain used for the training process (45–263 °C).

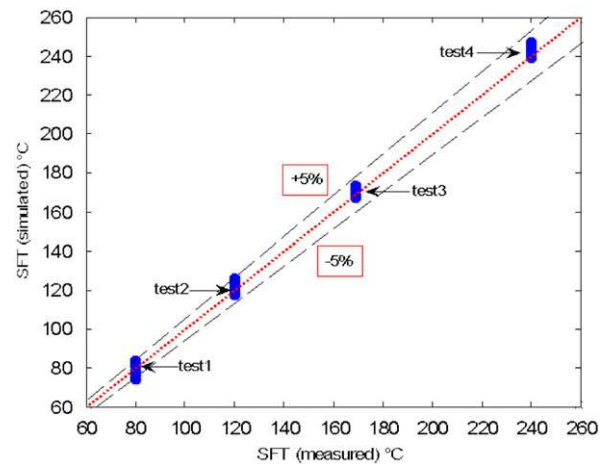


Fig. 7. Evaluation of ANN accuracy by using a statistical comparison between ANN simulated SFT values and true SFT data or TFT (reported in synthetic and actual borehole data sets).

Other additional advantages of the ANN model included: (i) to use the ANN tool without any assumptions about the nature of underlying mechanisms; (ii) to have the ability to take into account non-linearities and iterations between input and output variables; (iii) to optimise the computing time because these models rely on simple mathematical operations (allowing faster calculations); and (iv) to obtain reliable and faster predictions of SFT in comparison with those inferred from current analytical methods.

4. Conclusions

A new ANN model was successfully developed to predict the SFT in geothermal boreholes. It only requires BHT and shut-in time measurements as main input data. This feature makes it an advantageous tool over current available analytical methods that strongly depend on other complex input variables, such as the drilling mud circulation time. The ANN model was effectively trained with an experimental database, and validated with an unbiased testing database. The validity of the SFT computed by the ANN model was confirmed through a statistical comparison between the measured and simulated values, showing a good agreement with errors less than $\pm 5\%$.

The new ANN model can be used as a practical tool for the reliable determination of SFT in geothermal boreholes, especially when the BHT data are in the training interval (45 to ~ 263 °C). The ANN application interval could be extended by using higher BHT measurements. The predicting accuracy of the ANN could also be improved by using training patterns with a large number of BHT measurements, and their uncertainties.

The idea of applying ANN for the estimation of SFT is to propose a new practical and faster tool for the geothermal industry. In fact, the ANN model here developed could be coupled with smart BHT sensors for on-line determination of SFT in geothermal boreholes.

Finally, the ANN model developed may enable the implementation of smart sensors for on-line determination (SFT) in geothermal boreholes.

Acknowledgements

The authors wish to thank to the Engineering Ph.D. Programme of UNAM and to CONACyT for the financial support provided.

Thanks are also due to Dr. A. Santoyo-Castelazo (University of Liverpool) for her valuable comments. We are grateful to the anonymous reviewer, Dr. Viacheslav V. Spichak, and Prof. Dr. E.C. Grunsky (Editor in Chief) for their helpful comments on an earlier version of this paper.

References

- Andaverde, J., Verma, S.P., Santoyo, E., 2005. Uncertainty estimates of static formation temperature in borehole and evaluation of regression models. *Geophysical Journal International* 160, 1112–1122.
- Ascencio, F., García, A., Rivera, J., Arellano, V., 1994. Estimation of undisturbed formation temperatures under spherical radial heat flow conditions. *Geothermics* 23, 317–326.
- Bayram, A.F., 2001. Application of an artificial neural network model to a Na/K geothermometer. *Journal of Volcanology and Geothermal Research* 112, 75–81.
- Brennand, A.W., 1984. A new method for the analysis of static formation temperature test. In: *Proceedings of the 6th New Zealand Geothermal Workshop*, Auckland, New Zealand, pp. 45–47.
- Can, I., 2002. A new improved Na/K geothermometer by artificial neural networks. *Geothermics* 31, 751–760.
- Cao, S., Hermanrud, C., Lerche, I., 1988a. Estimation of formation temperature from bottom-hole temperature measurements: COST #1 well, Norton Sound, Alaska. *Geophysics* 53, 1619–1621.
- Cao, S., Lerche, I., Hermanrud, C., 1988b. Formation temperature estimation by inversion of borehole measurements. *Geophysics* 53, 979–988.
- Davis, A.P., Michaelides, E.E., 2009. Geothermal power production from abandoned oil wells. *Energy* 34, 866–872.
- Da-Xin, L., 1986. Non-linear fitting method of finding equilibrium temperature from BHT data. *Geothermics* 15, 657–664.
- Demuth, H., Beale, M., 1998. *Neural Network Toolbox for Matlab-User's Guide Version 3*. The MathWorks Inc., Natick, MA 742 pp.
- Díaz-González, L., Santoyo, E., Reyes, J., 2008. Tres nuevos geotermómetros mejorados de Na/K usando herramientas computacionales y geoquimiométricas: aplicación a la predicción de temperaturas de sistemas geotérmicos ("Three new improved geothermometers of Na/K using computational and geochemometric tools: application to the prediction of temperatures in geothermal systems"). *Revista Mexicana de Ciencias Geológicas* 25, 465–482.
- Dowdle, W.L., Cobb, W.M., 1975. Static formation temperature from well logs—an empirical method. *Journal of Petroleum Technology* 11, 1326–1330.
- Espinosa, G., García, A., Santoyo, E., Hernández, I., 2001. TEMLOPI/V.2: a computer program for estimation of fully transient temperatures in geothermal wells during circulation and shut-in. *Computers & Geosciences* 27, 327–344.
- Espinosa-Paredes, G., Espinosa-Martínez, E.G., 2009. A feedback-based inverse heat transfer method to estimate unperturbed temperatures in wellbores. *Energy Conversion and Management* 50, 140–148.
- Farshad, F.F., Garber, J.D., Lorde, J.L., 2000. Predicting temperature profiles in producing oil wells using artificial neural networks. *Engineering Computations* 17, 735–754.
- Fomin, S., Chugunov, V., Hashida, T., 2003. Analytical modelling of the formation temperature stabilization during the borehole shut-in period. *Geophysical Journal International* 155, 469–478.
- Fomin, S., Hashida, T., Chugunov, V., Kuznetsov, A.V., 2005. A borehole temperature during drilling in a fractured rock formation. *International Journal of Heat and Mass Transfer* 48, 385–394.
- González-Partida, E., García-Gutiérrez, A., Torres-Rodríguez, V., 1997. Thermal and petrologic study of the CH-A well from the Chipilapa-Ahuachapan geothermal area, El Salvador. *Geothermics* 26, 701–713.
- Goutorbe, B., Lucazeau, F., Bonneville, A., 2006. Using neural networks to predict thermal conductivity from geophysical well logs. *Geophysical Journal International* 166, 115–125.
- Hasan, A.R., Kabir, C.S., 1994. Static reservoir temperature determination from transient data after mud circulation. *Society of Petroleum Engineering, Drilling & Completion* 9, 17–24.
- Hagan, M.T., Menhaj, M., 1994. Training feed forward networks with the Marquardt algorithm. *IEEE Transactions on Neural Networks* 5, 989–993.
- Hermanrud, C., Cao, S., Lerche, I., 1990. Estimates of virgin rock temperature derived from BHT measurements: bias and error. *Geophysics* 55, 924–931.
- Hernández-Pérez, J.A., García-Alvarado, M.A., Trystram, G., Heyd, B., 2004. Neural networks for the heat and mass transfer prediction during drying of cassava and mango. *Food Science and Emerging Technologies* 5, 57–64.
- Hsieh, B.Z., Wang, C.W., Lin, Z.S., 2009. Estimation of formation strength index of aquifer from neural networks. *Computers & Geosciences* 35, 1933–1939.
- Hyodo, M., Takasugi, S., 1995. Evaluation of the curve-fitting method and the Horner plot for estimation of the true formation temperature using temperature recovery logging data. In: *Proceedings of the 20th Workshop on Geothermal Reservoir Engineering*. Stanford University, Stanford, USA, pp. 94–100.
- Iglesias, E.R., Campos-Romero, A., Torres, R.J., 1995. A reservoir engineering assessment of the Chipilapa, El Salvador, geothermal field. In: *Proceedings of the World Geothermal Congress*, Florence, Italy, pp. 1531–1536.
- Kutasov, I.M., 1999. *Applied Geothermics for Petroleum Engineers*. Elsevier, Amsterdam 360 pp.
- Kutasov, I.M., Eppelbaum, L.V., 2003. Prediction of formation temperatures in permafrost regions from temperature logs in deep wells-field cases. *Permafrost and Periglacial Process* 14, 247–258.
- Kutasov, I.M., Eppelbaum, L.V., 2005. Determination of formation temperature from bottom-hole temperature logs—a generalized Horner method. *Journal of Geophysical Engineering* 2, 90–96.
- Leblanc, Y., Pascoe, L.J., Jones, F.W., 1981. The temperature stabilization of a borehole. *Geophysics* 46, 1301–1303.
- Leite, E.P., Filho, C.R.de S., 2009a. Probabilistic neural networks applied to mineral potential mapping for platinum group elements in the Serra Leste region, Carajás mineral province, Brazil. *Computers & Geosciences* 35, 675–687.
- Leite, E.P., Filho, C.R.de S., 2009b. TEXTNN—a MATLAB program for textural classification using neural networks. *Computers & Geosciences* 35, 2084–2094.
- Limin, F., 1995. *Neural Networks in Computer Intelligence*. McGraw-Hill, New York 460 pp.
- Manetti, G., 1973. Attainment of temperature equilibrium in holes during drilling. *Geothermics* 2, 94–100.
- Morton, J.C., 2009. Boosting a fast neural network for supervised land cover classification. *Computers & Geosciences* 35, 1280–1295.
- Rumelhart, D.E., Hinton, G.E., Williams, R.J., 1986. Learning internal representations by error propagation. *Parallel Distributed Processing* 1, 318–362.
- Saito, S., Sakuma, S., Uchida, T., 1998. Drilling procedures, techniques and test results for a 3.7 km deep, 500 °C exploration well, Kakkonda, Japan. *Geothermics* 27, 573–590.
- Santoyo, E., García, A., Espinosa, G., Hernández, I., Santoyo, S., 2000. STATIC_TEMP: a useful computer code for calculating static formation temperatures in geothermal wells. *Computers & Geosciences* 26, 201–217.
- Schoepfel, R.J., Gilarranz, S., 1966. Use of well log temperatures to evaluate regional geothermal gradients. *Journal of Petroleum Technology* 18, 667–673.
- Serpen, G., Palabiyik, Y., Serpen, U., 2009. An artificial neural network model for Na/K geothermometer. In: *Proceedings of the 34th Workshop on Geothermal Reservoir Engineering*. Stanford University, Stanford, USA, pp. 1–12.
- Shen, P.Y., Beck, A.E., 1986. Stabilization of bottomhole temperature with finite circulation time and fluid flow. *Geophysical Journal of the Royal Astronomical Society* 86, 63–90.
- Spichak, V.V., 2006. Estimating temperature distributions in geothermal areas using a neuronet approach. *Geothermics* 35, 181–197.
- Spichak, V.V., Goidina, A.G., 2005. Temperature prediction in geothermal zones from borehole measurements using neural networks. *Physics of the Solid Earth* 4, 844–852.
- Steingrímsson, B., Gudmundsson, A., 2006. Geothermal borehole investigations during and after drilling. In: *Proceedings at Workshop on Decision Makers on Geothermal Projects in Central America*, El Salvador, pp. 1–10.
- Verma, S.P., 2005. *Estadística Básica para el Manejo de Datos Experimentales con la Aplicación en la Geoquímica—Geoquimiometría (Basic Statistics for Experimental Data Reduction with Application in Geochemistry—Geochemometrics)*. Universidad Nacional Autónoma de México, México, DF 186 pp.
- Verma, S.P., Andaverde, J., Santoyo, E., 2006a. Application of the error propagation theory in estimates of static formation temperatures in geothermal and petroleum boreholes. *Energy Conversion and Management* 47, 3659–3671.
- Verma, S.P., Andaverde, J., Santoyo, E., 2006b. Statistical evaluation of methods for the calculation of static formation temperatures in geothermal and oil wells using an extension of the error propagation theory. *Journal of Geochemical Exploration* 89, 398–404.
- Verma, S.P., Díaz-González, L., González-Ramírez, R., 2009. Relative efficiency of single-outlier discordancy tests for processing geochemical data on reference materials. *Geostandards and Geoanalytical Research* 33, 29–49.
- Wisian, K.W., Blackwell, D.D., Bellani, S., Henfling, J.A., Normann, R.A., Lysne, P.C., Forster, A., Schrotter, J., 1998. Field comparison of conventional and new technology temperature logging systems. *Geothermics* 27, 131–141.
- Yilmaz, S., Demircioglu, C., Akin, S., 2002. Application of artificial neural networks to optimum bit selection. *Computers & Geosciences* 28, 261–269.
- Zhang, G., Patuwo, B.E., Hu, M.Y., 1998. Forecasting with artificial neural networks: the state of the art. *International Journal of Forecasting* 14, 35–62.